Exact-ledger decomputerization: Packets A and B

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Abstract

This note replaces two groups of executable exact-arithmetic checks in the three-dimensional spherical-shell Pólya proof. Packet A proves once and for all the strict-bracket convention, odd-multiplicity block sums, the optical product-deficit identity, and monotonicity of propagated zero and channel cuts. Packet B gives exact hand-checkable tables for the five moving $3z$-crossings and for the seven first-excluded checkpoint channels. It then proves, without a finite search, that the five printed checkpoint inventories are exhaustive. Every numerical comparison below is an equality of rational numbers, a cross multiplication, or a positive-side squaring. No program output, decimal approximation, or interval calculation is a premise.

Contents

1 Scope and notation

Throughout $0 < \rho < 1$. Put

$$z = z_\rho := \frac{\pi}{1-\rho}, \qquad k_m(\rho)^2 := z_\rho^2 + m(m+1),$$

and define the two ratio endpoints

$$\rho_0 := \frac{1}{\sqrt{337}}, \qquad \rho_c := \frac{1}{1+2\pi}.$$

The high-checkpoint argument uses the strict proxy

$$b_{\ell,n}(z)^2 := \max\{n^2z^2 + \ell(\ell+1),\ \beta_{\ell,n}^2\}. \tag{1}$$

A channel may be counted at K only when $b_{\ell,n}(z) < K$. The constants $\beta_{\ell,n}$ are strict lower bounds for the corresponding ball zeros. The zero bounds used in the checkpoint table are displayed explicitly in Section 5.

The purpose of this note is deliberately limited. It replaces the general exact-arithmetic and checkpoint-ordering rows of the old ledger; it does not replace the zero-sign proofs, high-event payment registry, seam arithmetic, optical endpoint screens, or formal owner subtraction.

2 A rational enclosure for π

Lemma 2.1 (Hand enclosure for π)**.** *One has*

$$\frac{333}{106} < \pi < \frac{22}{7}, \qquad \frac{1}{\pi} < \frac{106}{333}. \tag{2}$$

Proof. For $0 < x \le 1$, the alternating series for $\arctan x$ gives a strict lower bound after a negative term and a strict upper bound after a positive term. Hence

$$\arctan\frac{1}{5} > \frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5} - \frac{1}{7\cdot 5^7} = \frac{323852}{1640625}, \qquad \arctan\frac{1}{239} < \frac{1}{239}.$$

Machin's identity therefore yields

$$\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239} > \frac{1231847548}{392109375},$$

and direct subtraction gives the positive margin

$$\frac{1231847548}{392109375} - \frac{333}{106} = \frac{3418213}{41563593750} > 0.$$

For the upper bound, polynomial division and elementary integration give

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

Finally, reciprocation of the positive lower bound for π gives the last inequality. Notice also that $22/7 - 333/106 = 1/742 > 0$, so the enclosure is nonempty. $\square$

3 Packet A: universal exact-arithmetic lemmas

3.1 Strict floors and equality owners

Definition 3.1. For $x \in \mathbb{R}$, let

$$[x]_< := \#\{q \in \mathbb{Z}_{\ge 1} : q < x\}.$$

Thus the bracket counts positive integers under a *strict* cutoff.

Lemma 3.2 (Strict-bracket identity). *For every real x,*

$$[x]_< = \max\{0, \lceil x \rceil - 1\}. \tag{3}$$

In particular, for every integer $m \geq 1$,

$$[m]_< = m - 1, \qquad [m + \tfrac{1}{2}]_< = m.$$

Consequently an integer equality wall is owned by the excluded side.

Proof. If $x \leq 1$, both sides of (3) vanish. If $x > 1$, the positive integers below x are exactly $1, \ldots, \lceil x \rceil - 1$. This also proves the two specializations. □

The same lemma handles the two equality conventions used later. If $A + 1/4 = n \in \mathbb{Z}_{\geq 1}$, then $[A + 1/4]_< = n - 1$. If $a = m\pi$ in the optical radial decomposition, the strict radial layers are represented by $(N, t) = (m - 1, 1)$, not by $(m, 0)$.

3.2 Odd multiplicities and block caps

Lemma 3.3 (Odd-sum and block-cap identity). *For every integer $h \geq 0$,*

$$\sum_{\ell=0}^{h}(2\ell + 1) = (h + 1)^2, \qquad \sum_{\ell=0}^{h-1}(2\ell + 1) = h^2. \tag{4}$$

More generally, if radial layer n contains a complete angular block $0 \leq \ell \leq h_n$, with $h_n = -1$ denoting an empty block, then its total multiplicity is

$$\sum_n (h_n + 1)^2. \tag{5}$$

If $x^2 = h(h + 1)$ is a strict product equality wall, the equality channel $\ell = h$ is absent and the contribution is h^2.

Proof. The first identity follows either by induction or by $(\ell + 1)^2 - \ell^2 = 2\ell + 1$. Summing over independent radial layers gives (5). At $x^2 = h(h + 1)$, $\sqrt{x^2 + 1/4} - 1/2 = h$; Lemma 3.2 excludes $\ell = h$, leaving $0 \leq \ell < h$ and hence h^2. □

For completeness, the cap arithmetic collapsed by this lemma is printed in compact form below. A dash means that another analytic argument has declared the cell impossible; the lemma asserts only the arithmetic of the non-dash entries.

$k_8 : H\backslash h$	$\varnothing$	0	1	2	3
7	64	65	68	73	80
6	49	50	53	–	–
5	36	37	40	45	52
4	25	26	29	34	41

$k_9 : H\backslash h$	$\varnothing$	0	1	2	3	4
8	81	–	–	–	–	–
7	64	65	68	73	–	–
6	49	50	53	58	65	74
5	36	37	40	45	52	61
4	25	26	29	34	41	50

Indeed every entry is $(H+1)^2$ plus 0 or $(h+1)^2$. At k_7, the first blocks $H = 4, 5, 6$ give $25, 36, 49$, and second channels $0, 1$ contribute $1 + 3 = 4$. The remaining printed caps decompose as

$$\begin{aligned} k_{10}: \quad & 10^2; \quad 9^2,\ 9^2+4^2; \\ & 8^2,\ 8^2+4^2,\ 8^2+5^2,\ 8^2+6^2; \\ & 7^2+5^2,\ 7^2+4^2+2^2,\ 7^2+5^2+2^2, 7^2+6^2,\ 7^2+6^2+2^2 \\ & = 100; \quad 81, 97; \quad 64, 80, 89, 100; \quad 74, 69, 78, 85, 89, \end{aligned}$$

$$\begin{aligned} k_{11}: \quad & 11^2; \quad 10^2,\ 10^2+5^2; \quad 9^2+5^2,\ 9^2+5^2+2^2; \quad 8^2+5^2+2^2 \\ & = 121; \quad 100, 125; \quad 106, 110; \quad 93. \end{aligned}$$

The conservative cap-74 row is the same identity $7^2 + 5^2 = 74$.

The six lower inventories similarly give, in their printed order,

$$(6^2+3^2,\ 6^2+3^2+1,\ 7^2+3^2+1, 7^2+4^2+1,\ 7^2+4^2+1+3,\ 7^2+5^2+1+3) = (45, 46, 59, 66, 69, 78).$$

Finally, the ten lower-d caps are the successive odd sums

channels	cap	channels	cap
1	1	$1+3$	4
$1+3+5$	9	$1+3+5+1$	10
$1+3+5+7$	16	$1+3+5+7+1$	17
$1+3+5+7+9+1$	26	$1+3+5+7+9+1+3$	29
$1+3+5+7+9+11+1+3$	40	$1+3+5+7+9+11+1+3+5$	45.

3.3 The symbolic optical product deficit

Proposition 3.4 (Product-deficit identity and uniform reserve)**.** *Let $N \geq 1$ be an integer, $0 < t \leq 1$, and $x = N + t$. Then*

$$\frac{2}{3}x^3 - Nx^2 + \sum_{n=1}^{N} n^2 = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{6}$$

For $C_D = 1382/3125$, the right-hand side is strictly larger than $C_D(N+t)^2$. More precisely, its difference from $C_D(N+t)^2$ is at least

$$\frac{1822532}{91552734375} > 0. \tag{7}$$

Proof. Insert $\sum_{n=1}^{N} n^2 = N(N+1)(2N+1)/6$ and expand; this proves (6) for all N, t, rather than at sampled values. Put $C = C_D$, $A = 1/2 - C = 361/6250$, and let $G_N(t)$ denote the difference under consideration. Then

$$G_N(t) = AN^2 + N\left(t^2 - 2Ct + \frac{1}{6}\right) + \frac{2t^3}{3} - Ct^2.$$

Consequently

$$G_{N+1}(t) - G_N(t) = A(2N+1) + t^2 - 2Ct + \frac{1}{6}.$$

Since $0 < C < 1$, the last quadratic has minimum $1/6 - C^2$ on $(0, 1]$. For $N \geq 1$,

$$G_{N+1}(t) - G_N(t) \geq 3A + \frac{1}{6} - C^2 = \frac{4229603}{29296875} > 0.$$

It remains to minimize G_1. Direct differentiation gives

$$G_1'(t) = 2(t - C)(t + 1),$$

so its unique minimum on $0 < t \leq 1$ is at $t = C$. Exact substitution gives

$$G_1(C) = \frac{1822532}{91552734375} > 0,$$

which proves both assertions. $\square$

3.4 Monotonic propagated cuts

Lemma 3.5 (Zero-cut/channel-cut monotonicity)**.** *Fix $n \geq 2$, a checkpoint m, and $M = m(m+1)$. Let $\mathcal{A}_n$ be a finite collection of anchor pairs $(j, q_{j,n}^2)$, where $j_{j+1/2,n}^2 > q_{j,n}^2$ at each anchor. For every ℓ above at least one anchor, set*

$$\mathfrak{B}_n(\ell) := \max_{(j,q_{j,n}^2)\in\mathcal{A}_n,\ j\leq\ell} \left(q_{j,n}^2 + \ell(\ell+1) - j(j+1)\right), \tag{8}$$

$$\mathfrak{Z}_{m,n}(\ell) := \mathfrak{B}_n(\ell) - M, \tag{9}$$

$$\mathfrak{C}_{m,n}(\ell) := \frac{M - \ell(\ell+1)}{n^2 - 1}. \tag{10}$$

Then

$$\mathfrak{Z}_{m,n}(\ell+1) \geq \mathfrak{Z}_{m,n}(\ell) + 2(\ell+1) > \mathfrak{Z}_{m,n}(\ell), \tag{11}$$

$$\mathfrak{C}_{m,n}(\ell+1) = \mathfrak{C}_{m,n}(\ell) - \frac{2(\ell+1)}{n^2-1} < \mathfrak{C}_{m,n}(\ell). \tag{12}$$

In particular, if $\mathfrak{Z}_{m,n}(\ell_0) \geq \mathfrak{C}_{m,n}(\ell_0)$, then no channel (ℓ, n) with $\ell \geq \ell_0$ can satisfy the strict proxy (1) *at $K = k_m$.*

Proof. Every old candidate in (8) gains exactly $(\ell+1)(\ell+2) - \ell(\ell+1) = 2(\ell+1)$ when ℓ is replaced by $\ell + 1$; a new anchor can only increase the maximum. This proves (11). Formula (12) is direct.

If a channel were counted at k_m, its product part would give

$$(n^2 - 1)z^2 < M - \ell(\ell+1), \quad \text{hence} \quad z^2 < \mathfrak{C}_{m,n}(\ell),$$

whereas its propagated ball-zero part would give

$$\mathfrak{B}_n(\ell) < z^2 + M, \quad \text{hence} \quad z^2 > \mathfrak{Z}_{m,n}(\ell).$$

These strict inequalities are incompatible when $\mathfrak{Z} \geq \mathfrak{C}$. Equations (11) and (12) propagate the incompatibility to every larger angular index. $\square$

4 Packet B: moving-$3z$ ordering

We first record rational radical enclosures. All six follow by positive squaring; the exact square margins are

lower comparison	margin	upper comparison	margin
$(183/10)^2 < 337$	211/100	$337 < (92/5)^2$	39/25
$(1051/100)^2 < 7073/64$	2221/40000	$7073/64 < (1052/100)^2$	6191/40000
$(1067/100)^2 < 114$	1511/10000	$114 < (107/10)^2$	49/100.

Thus

$$\frac{183}{10} < \sqrt{337} < \frac{92}{5}, \quad \frac{1051}{100} < \frac{\sqrt{7073}}{8} < \frac{1052}{100}, \quad \frac{1067}{100} < \sqrt{114} < \frac{107}{10}. \tag{13}$$

Lemma 4.1 (Endpoint and fixed-wall order)**.** *With* $d = \sqrt{337}/2$*,*

$$d < 3z_{\rho_0} < 10 < \frac{81}{8} < \frac{21}{2} < \frac{\sqrt{7073}}{8} < \sqrt{114} < 3z_{\rho_c}. \tag{14}$$

Moreover $(81/8)^2 + 8 = 7073/64$.

Proof. The nontrivial endpoint estimates have the following rational margins. From (2) and (13),

$$3z_{\rho_0} > 3\pi > \frac{999}{106}, \qquad d < \frac{46}{5}, \qquad \frac{999}{106} - \frac{46}{5} = \frac{119}{530} > 0.$$

Also $\rho_0 < 10/183$, whence

$$3z_{\rho_0} < \frac{66}{7}\frac{183}{173} = \frac{12078}{1211}, \qquad 10 - \frac{12078}{1211} = \frac{32}{1211} > 0.$$

At the other endpoint, $z_{\rho_c} = \pi + 1/2$; therefore

$$3z_{\rho_c} > \frac{579}{53}, \qquad \sqrt{114} < \frac{107}{10}, \qquad \frac{579}{53} - \frac{107}{10} = \frac{119}{530} > 0.$$

The rational middle comparisons are immediate. For the two radical steps one may square positive quantities:

$$\frac{7073}{64} - \left(\frac{21}{2}\right)^2 = \frac{17}{64} > 0, \qquad 114 - \frac{7073}{64} = \frac{223}{64} > 0.$$

Finally, $(81/8)^2 + 8 = (6561 + 512)/64 = 7073/64$. $\square$

For a positive wall w, the unique solution of $3z_\rho = w$ is

$$r_w := 1 - \frac{3\pi}{w}.$$

The following table encloses each crossing between explicit rational numbers. The bounds use only Lemma 2.1 and (13); for instance r_w increases with w and decreases with π.

wall w	exact rational enclosure for w	exact enclosure for r_w
10	10	$\frac{2}{35} < r_{10} < \frac{61}{1060}$
81/8	81/8	$\frac{13}{189} < r_{81/8} < \frac{11}{159}$
21/2	21/2	$\frac{5}{49} < r_{21/2} < \frac{38}{371}$
$\sqrt{7073}/8$	$1051/100 < w < 1052/100$	$\frac{757}{7357} < r_w < \frac{2903}{27878}$
$\sqrt{114}$	$1067/100 < w < 107/10$	$\frac{79}{679} < r_w < \frac{676}{5671}$

The four gaps between consecutive rational enclosures are, in order,

$$\frac{2251}{200340}, \qquad \frac{256}{7791}, \qquad \frac{183}{389921}, \qquad \frac{231225}{18929162}, \tag{15}$$

and are all positive. For example, the first is $13/189 - 61/1060$, and similarly for the other three. The endpoint margins are also explicit:

$$\rho_0 < \frac{10}{183} < \frac{2}{35}, \quad \frac{2}{35} - \frac{10}{183} = \frac{16}{6405} > 0,$$

and, since $\rho_c > 7/51$,

$$\frac{676}{5671} < \frac{7}{51} < \rho_c, \quad \frac{7}{51} - \frac{676}{5671} = \frac{5221}{289221} > 0.$$

We have therefore proved the complete strict order

$$\rho_0 < r_{10} < r_{81/8} < r_{21/2} < r_{\sqrt{7073}/8} < r_{\sqrt{114}} < \rho_c. \tag{16}$$

This single table replaces the ten crossing-location predicates and the four crossing-order predicates of the executable registry.

5 Packet B: checkpoint boundary table and exhaustiveness

The ball-zero input consists of the following strict rational lower bounds:

$$\begin{array}{c|cccc} n=2,\ \ell & 1 & 2 & 3 & 5 \\ \hline q_{\ell,2}^2 & (77/10)^2 & 9^2 & (103/10)^2 & (129/10)^2 \end{array} \qquad \begin{array}{c|cc} n=3,\ \ell & 1 & 2 \\ \hline q_{\ell,3}^2 & (21/2)^2 & (61/5)^2. \end{array} \tag{17}$$

Angular propagation gives the lower bound (8) at all larger angular indices. The seven first-excluded channels have the exact cuts shown below. Here $M = m(m+1)$, $B = \mathfrak{B}_n(\ell_0)$, $Z = B - M$, $C = (M - \ell_0(\ell_0+1))/(n^2-1)$, and the last column is the decisive incompatibility margin $Z - C$.

m	n	first excluded ℓ_0	B	Z	C	$Z - C$
7	2	2	81	25	50/3	25/3
8	2	4	11409/100	4209/100	52/3	7427/300
9	2	5	16641/100	7641/100	20	5641/100
10	2	6	17841/100	6841/100	68/3	13723/300
11	2	5	16641/100	3441/100	34	41/100
10	3	2	3721/25	971/25	13	646/25
11	3	2	3721/25	421/25	63/4	109/100

Every table entry is reconstructible from (17). The only propagated base values not already anchors are

$$\frac{11409}{100} = \left(\frac{103}{10}\right)^2 + (4\cdot 5 - 3\cdot 4), \qquad \frac{17841}{100} = \left(\frac{129}{10}\right)^2 + (6\cdot 7 - 5\cdot 6).$$

All seven margins are positive; the smallest is $41/100$, so none of these exclusions relies on an equality convention. By Lemma 3.5, each row excludes not merely the displayed channel but its entire angular tail.

It remains to rule out the radial tails. On the high strip $\rho_c \le \rho < 1$,

$$z \ge z_{\rho_c} = \pi + \frac{1}{2} > \frac{333}{106} + \frac{1}{2} = \frac{193}{53}, \qquad z^2 > \frac{37249}{2809}.$$

The exact margins needed below are

$$\frac{37249}{2809} - \frac{90}{8} = \frac{22591}{11236} > 0, \qquad \frac{37249}{2809} - \frac{132}{15} = \frac{62649}{14045} > 0. \tag{18}$$

At $m \le 9$, an $n = 3$ channel would require $8z^2 < M - \ell(\ell+1) \le 90$, contradicting the first inequality. At $m \le 11$, every $n \ge 4$ channel would require $15z^2 \le (n^2-1)z^2 < M - \ell(\ell+1) \le 132$, contradicting the second inequality. These are the two base radial exclusions.

Theorem 5.1 (Exhaustive checkpoint inventories)**.** *For $\rho_c \le \rho < 1$, the channels that can satisfy the strict proxy* (1) *at the five checkpoints are contained in the following lists:*

checkpoint	*possible modes*
k_7	*first* 0:6*, second* 0:1*, no* $n \ge 3$
k_8	*first* 0:7*, second* 0:3*, no* $n \ge 3$
k_9	*first* 0:8*, second* 0:4*, no* $n \ge 3$
k_{10}	*first* 0:9*, second* 0:5*, third* 0:1*, no* $n \ge 4$
k_{11}	*first* 0:10*, second* 0:4*, third* 0:1*, no* $n \ge 4$*.*

Here n-th means radial index n, and a:b denotes the angular indices $a, a+1, \ldots, b$. The word "possible" asserts an exhaustive upper inventory, not that every listed channel is present for every ratio.

Proof. For $n = 1$, the product part of the strict proxy at k_m is

$$z^2 + \ell(\ell+1) < z^2 + m(m+1).$$

Thus $\ell < m$. The equality channel $\ell = m$ and all larger indices are excluded, with the equality ownership fixed by Lemma 3.2. This gives precisely first 0:$m-1$.

For $n = 2$, apply the first five rows of the checkpoint table and then Lemma 3.5. They give first-excluded angular indices $2, 4, 5, 6, 5$ at $m = 7, 8, 9, 10, 11$, respectively. Hence the second blocks end at $1, 3, 4, 5, 4$.

For $n = 3$, the first inequality in (18) removes all channels through k_9. At k_{10} and k_{11}, the last two rows of the checkpoint table and cut monotonicity remove every $\ell \ge 2$, leaving only $\ell = 0, 1$. Finally, the second inequality in (18) removes all $n \ge 4$ through k_{11}. The five lists follow, without any bounded angular search. ☐

6 Executable-row coverage contract

The former executable baseline contained 611 rows and had canonical digest

`afd7e054f52aa9a3992fc8ef16d3e7551586c76bfb589053714f5f55a36792f1`.

The digest is included only to identify the audited baseline. It is not a premise of any proof above. The exhibits above replace disjoint row groups as follows:

analytic exhibit	rows	exact ledger subfamilies
Lemma 2.1	4	one enclosure-nonemptiness row, two rational π-bound rows, and one reciprocal row
Lemma 3.2	24	sixteen integer/half-integer strict brackets, seven phase-proxy integer walls, and one optical radial equality owner
Lemma 3.3	97	twelve strict angular equality blocks, twelve optical angular blocks, 62 high-checkpoint cap identities, one six-cap lower tuple, and ten lower-d cap identities
Proposition 3.4	23	five general product-deficit predicates and eighteen sampled expansions
Lemma 3.5, the checkpoint boundary table, and Theorem 5.1	117	76 repeated cut-monotonicity rows, seven first-excluded comparisons, two radial exclusions, and 32 metadata/tail instances
Lemma 4.1 and the moving-crossing table	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, ten crossing locations, and four crossing-order rows
total	**288**	

The following family table gives the same accounting against the executable entry points and makes the partial-family boundaries explicit. It maps the exact rows replaced by this note. "Collapsed" means that multiple sampled or bookkeeping rows are consequences of one displayed identity; it does not mean that an unprinted program enumeration is being used.

executable family	family rows	replaced here	analytic replacement
`verify_pi`	4	4	Lemma 2.1
`verify_thresholds`	34	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, and fourteen moving-crossing predicates
`verify_strict_face_conventions`	35	35	Lemmas 3.2 and 3.3
`verify_lower_inventories_and_payments`	44	1	single six-cap tuple bookkeeping row
`verify_checkpoint_inventories`	117	117	boundary table, radial margins, and Theorem 5.1
`verify_cap_tables_and_payments`	98	62	all block-cap arithmetic: 61 cap-table rows plus the cap-74 block identity
`verify_optical_constants`	50	36	twelve angular blocks, one radial equality owner, five general product-deficit facts, and eighteen sampled expansions
supplemental k_6/lower-d ledger	26	10	ten lower-d block-cap identities
total		**288**	

For the checkpoint family, the 117 rows decompose exactly as follows: 32 finite metadata/tail instances, seven first-excluded zero/channel comparisons, two base radial exclusions, and 76 repeated cut-monotonicity instances. Theorem 5.1 replaces all four groups. For the optical family, Proposition 3.4 replaces the 18 sampled (N, t) expansions by a symbolic identity valid for every integer $N \geq 1$ and every $0 < t \leq 1$.

The remaining 323 rows are outside Packets A and B. In particular, this note makes no claim to replace the high-event cap/payment implications, the Bessel sign registry, seam payments, optical screen constants, or the exact owner subtraction. Those require the other analytic replacement packets. This separation prevents a universal arithmetic lemma from being mistaken for proof of a geometric event assignment.